Energy Poverty and the Targeting of Electricity Subsidies: Evidence from the FOSE in Peru

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June 2026

Abstract

This paper evaluates whether the 140 kWh monthly consumption threshold used to determine eligibility for Peru's Electricity Social Compensation Fund (*Fondo de Compensación Social Eléctrica*, FOSE) adequately identifies energy-poor households. Using microdata from the 2025 Residential Energy Consumption and Use Survey (ERCUE, 14,815 households), I compare the incidence of energy poverty across households near the threshold using Coarsened Exact Matching (CEM) and inverse probability weighting (IPW), with two complementary indicators: energy overload (2M) and hidden energy poverty (HEP). The current threshold exhibits a large inclusion error (69.4%): on average there is no significant difference in energy poverty between comparable eligible and ineligible households (ATT $= -2.37$ pp, $p = 0.330$). However, among non-extreme poor households a sizeable effect emerges (-20.29 pp, $p = 0.014$), robust across specifications. Counterfactual simulations indicate that the FOSE plays a protective role in rural areas. The findings suggest that the consumption-based design is poorly targeted but retains positive effects in specific segments, pointing to the need to revise the eligibility criteria.

Keywords: energy poverty; FOSE; targeting; electricity subsidy; Peru; Coarsened Exact Matching.
JEL: Q41, Q48, H23, I38, L94.

1 Introduction

Energy poverty—a household's inability to access essential energy services at an affordable cost—is a policy problem that extends well beyond developed countries (EPOV, 2023). In Latin America, where electricity coverage approaches universality but affordability remains precarious, the phenomenon has received comparatively little empirical attention. Peru illustrates this tension: according to ERCUE 2025, 97% of households are connected to the grid, yet the median energy burden reaches 5.20% among extreme-poor households, and the gap between median urban (92 kWh/month) and rural (16 kWh/month) consumption reflects two radically different energy realities.

Peru's main direct consumer-protection instrument is the FOSE, a cross-subsidy created in 2001 that grants discounts on the electricity bill to households consuming below a regulated threshold—currently 140 kWh/month, raised from 100 kWh after the 2022 reform (D.S. 023-2022-PCM). Like many block-tariff subsidies, the FOSE uses electricity consumption as its sole eligibility criterion, implicitly assuming that low consumption signals economic vulnerability. This premise has been questioned across contexts. Whittington et al. (2015) show that consumption-based subsidies are poorly targeted toward the poorest, and Levinson & Silva (2022) document that the weak correlation between electricity consumption and income limits

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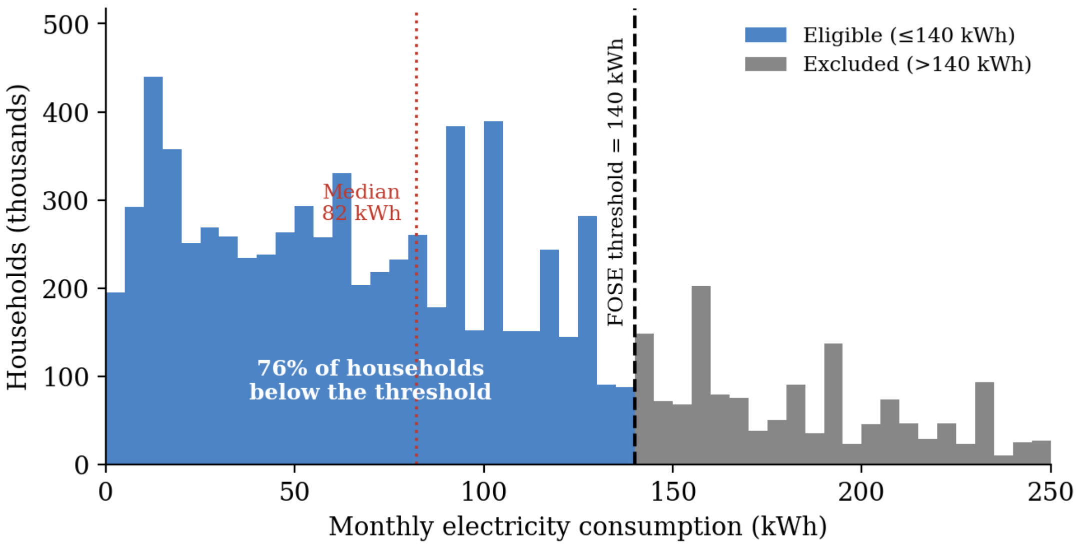

Figure 1: Weighted distribution of monthly electricity consumption and the FOSE eligibility threshold, ERCUE 2025. Source: ERCUE 2025; weighted by expansion factor; 5 kWh bins.

the redistributive capacity of tariff discounts. In Spain, evaluations of the analogous *Bono Social Eléctrico* find modest effects: a reduction of about two percentage points in the probability of energy poverty (Bagnoli & Bertoméu-Sánchez, 2022), no significant effect on subjective indicators (García Álvarez & Tol, 2021), and a declining effect that is nearly null for the poorest households (Llorca & Rodríguez-Álvarez, 2025).

Despite the FOSE's relevance for Peruvian social policy, no empirical test of its eligibility rule or of its effect on household-level energy-poverty indicators exists, to the best of my knowledge. This paper fills that gap. The central hypothesis is twofold. *First*, electricity consumption is an imperfect proxy for vulnerability: in the aggregate, the 140 kWh rule does not cleanly separate vulnerable from non-vulnerable households because the eligible group is heavily diluted by non-poor households; yet in the subgroup of non-extreme poor households the threshold does discriminate. *Second*, even where the rule points in the right direction, the size of the tariff benefit is too small to produce a clear separation in energy well-being around the threshold.

The analysis yields three main findings. The FOSE exhibits a 69.4% inclusion error and a normative exclusion pattern that disproportionately affects rural households with shared electricity connections. The 140 kWh threshold produces no significant separation in the probability of energy overload between comparable eligible and excluded households in the narrow ±20 kWh window, though the effect strengthens in wider windows and becomes large and significant once the sample is restricted to poor households. Finally, hidden energy poverty (HEP) is essentially zero in the neighbourhood of the threshold, revealing that energy poverty there manifests exclusively as financial overload rather than deprivation.

2 The FOSE and Peru's Energy Context

Peru has achieved near-universal electrification (97% of households). However, grid access guarantees neither adequate consumption nor an affordable price: the median energy burden is 3.13% nationally but reaches 5.20% among extreme-poor households. The distribution of consumption is bimodal by area (Figure 1): rural density concentrates below 30 kWh, while the urban distribution peaks at 60–90 kWh. Overall, 76% of households consume at or below the 140 kWh threshold and are therefore eligible for the FOSE.

Table 1: Electricity consumption distribution by typical sector, ERCUE 2025.

Sector	Mean (kWh)	Median (kWh)	≤30 (%)	31–140 (%)	>140 (%)	Poverty (%)
ST1 (Lima)	160.2	129.8	3.3	53.9	42.8	16.1
ST2 (Urban mid)	92.8	75.3	15.9	66.4	17.7	18.1
ST3 (Urban-rural)	52.7	36.8	43.0	50.1	6.9	34.8
ST4 (Rural)	32.6	17.0	67.7	29.3	3.0	32.9
SER (Rural isol.)	37.5	26.9	59.5	37.5	3.0	32.3

Source: ERCUE 2025.

The FOSE was created in 2001 (Law 27510) and operates as an intra-sectoral cross-subsidy: users with consumption above the threshold and free-market clients finance, through a surcharge, the discounts applied to beneficiaries. As of December 2024, two of every three residential users were FOSE beneficiaries, and the annual subsidy reached roughly 540 million soles (an 80% increase relative to 2019). The 2022 reform (Law 31429) raised the threshold and introduced additional exclusion criteria: no more than one property or supply point, not being located in a high- or middle-income block, and no consumption above 140 kWh during the summer months.

Heterogeneity by typical electricity sector is pronounced (Table 1). Lima (ST1) records average consumption of 160 kWh and 42.8% of households above the threshold, against the rural isolated systems (SER) with 37.5 kWh and barely 3%. The discount structure operates in three blocks: for the first 30 kWh the discount is proportional to the energy charge; between 31 and 140 kWh it is a fixed amount; above 140 kWh the household becomes a net contributor to the fund.

3 Data and Empirical Strategy

Data. The analysis uses microdata from ERCUE 2025, a probabilistic, stratified, multi-stage survey representative at the national, departmental and urban/rural level, with fieldwork conducted between September and November 2024. The sample comprises 15,485 households (14,815 with valid consumption data) across 25 departments, and the expansion factor scales the observations to the universe of 9.3 million households with electricity access. Two relevant limitations are that ERCUE does not contain the regulated tariff by electrical system (approximated through the implicit price, expenditure/consumption) nor the INEI block-stratum classification, and that the cross-sectional design precludes dynamic analysis.

Energy-poverty indicators. The household energy burden is $V_i = G_i^{\text{elec}} / G_i^{\text{total}}$. A household is classified as energy-overloaded (*2M*) when this ratio exceeds twice the national median:

$$2M_i = \mathbb{1}\left(V_i > 2\widetilde{V}\right). \tag{1}$$

Because indicators based solely on relative expenditure can understate deprivation when households suppress consumption, I add the hidden energy poverty (HEP) indicator, which flags households whose absolute electricity expenditure falls below half the median of their reference group (Thomson et al., 2016):

$$\text{HEP}_i = \mathbb{1}\left(G_i^{\text{elec}} < 0.5 \times \widetilde{G}^{\text{elec}}_{\text{group}_i}\right), \tag{2}$$

where the group is defined by geographic area.

Table 2: Targeting indicators by eligibility definition, ERCUE 2025.

Indicator	By consumption	Full rule (weighted)	Full rule (unweighted)
Definition	$C \leq 140$ kWh	+ meter + stratum	+ meter + stratum
Coverage	84.1%	76.1%	76.1%
Inclusion error	69.4%	75.5%	—
Exclusion error	5.3%	10.4%	15.2%

Source: ERCUE 2025.

Identification. The strategy compares households on either side of the 140 kWh administrative threshold within narrow consumption windows: a main specification of ±20 kWh ($N \approx 1{,}286$) and a robustness window of ±30 kWh ($N \approx 1{,}949$). The parameter of interest is $\tau = E[Y_i \mid C_i < 140] - E[Y_i \mid C_i \geq 140]$, with $Y_i \in \{2M_i, \mathrm{HEP}_i\}$; standard errors come from a nonparametric bootstrap with 2,000 replications. Although a regulatory threshold suggests a regression-discontinuity (RD) design, three features preclude its valid application: FOSE eligibility does not depend exclusively on consumption, the data are cross-sectional, and compositional heterogeneity around the threshold violates the continuity assumption.

Estimation. Coarsened Exact Matching (CEM, Iacus et al., 2012) is the main method. The final specification retains three covariates—grouped typical sector, area (urban/rural) and household size—generating 17 cells with 99.8% common support. The ATT estimator is $\hat{\tau}_{\mathrm{ATT}} = \sum_j (n_{1j}/N_1)(\overline{Y_{1j}} - \overline{Y_{0j}})$, where eligibility is $T_i = ⊮(C_i < 140)$. Inverse probability weighting (IPW) serves as a sensitivity check. This paper evaluates the discriminating capacity of the eligibility rule and should not be read as a causal estimate of the subsidy's impact on welfare.

4 Results

4.1 Targeting errors

84.1% of ERCUE 2025 households consume at or below 140 kWh, making them eligible for the FOSE. This near-universal coverage generates a 69.4% inclusion error: seven of every ten consumption-eligible households are not in monetary poverty. Disaggregation by sector shows the inclusion error is substantially larger in urban sectors (74.8% in ST1, 78.6% in ST2) than in rural sectors (60–63%), reflecting greater income heterogeneity in urban areas (Levinson & Silva, 2022). Exclusion error by the consumption criterion is 5.3% but rises to 20.3% in ST1. The main normative exclusion criterion in rural sectors is not the consumption threshold but the shared-meter condition: 8.5% of eligible SER households share an electricity supply point, an exclusion driven by agricultural arrangements rather than ability to pay.

Figure 2 shows that the 2022 reform pushed national coverage from 53.8% to 76.1% while the inclusion error edged up to 75.7% under the full rule. Table 2 summarises the targeting indicators under alternative eligibility definitions.

4.2 Discriminating power of the threshold

The crude difference between eligible and excluded households in the ±20 kWh window is −1.03 pp and not significant. Applying CEM, the ATT is −2.37 pp (SE = 2.43; $p = 0.330$). The ±30 kWh robustness window yields an ATT of −5.72 pp ($p = 0.003$), significant at the 1% level, and IPW gives a consistent −2.09 pp (SE = 2.35; $p = 0.376$). Common support is

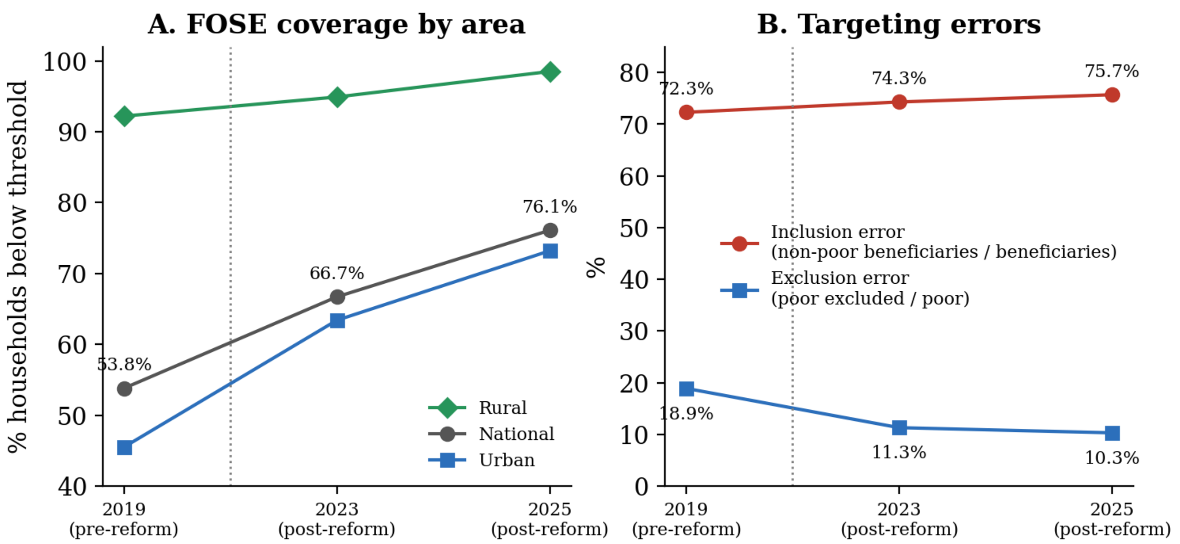

Figure 2: Evolution of FOSE coverage and targeting errors, 2019–2025. Source: own elaboration with ERCUE 2019, 2023 and 2025.

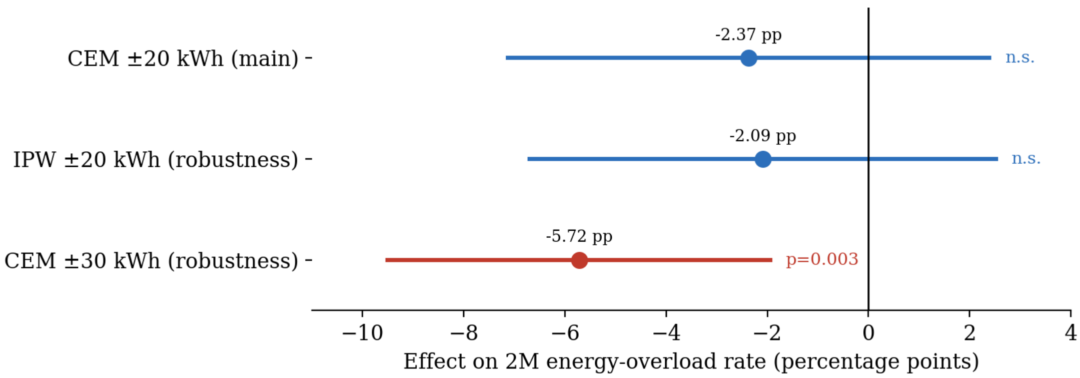

Figure 3: Difference in energy overload (2M) associated with FOSE eligibility: comparison of estimators. Source: ERCUE 2025.

adequate: the mean propensity score is 0.575 for eligibles and 0.571 for excluded households. Figure 3 compares the estimators.

The negative sign of the ATT and the significance in the wide window suggest the threshold points in the right direction, but the difference is modest and insignificant in the main window: the FOSE discount is too small to create a sharp separation. For a household consuming 130 kWh, the median FOSE discount is S/ 10.72, barely 10% of a S/ 105 bill—convergent with international evidence on tariff subsidies of insufficient scale (Bagnoli & Bertoméu-Sánchez, 2022; García Álvarez & Tol, 2021; Llorca & Rodríguez-Álvarez, 2025). Importantly, in the ±20 kWh window the HEP rate is exactly 0.0% for both eligible and excluded households: near the threshold, energy poverty manifests exclusively as financial overload, not deprivation.

4.3 Heterogeneity by poverty status

The most relevant finding emerges when disaggregating by monetary-poverty status (Figure 4). Among non-extreme poor households in the ±20 kWh window ($N = 121$), the ATT is -20.29 pp ($p = 0.014$). Among non-poor households ($N = 1{,}132$) the effect is essentially null. For extreme-poor households ($N = 9$) estimation is not feasible due to insufficient sample. The

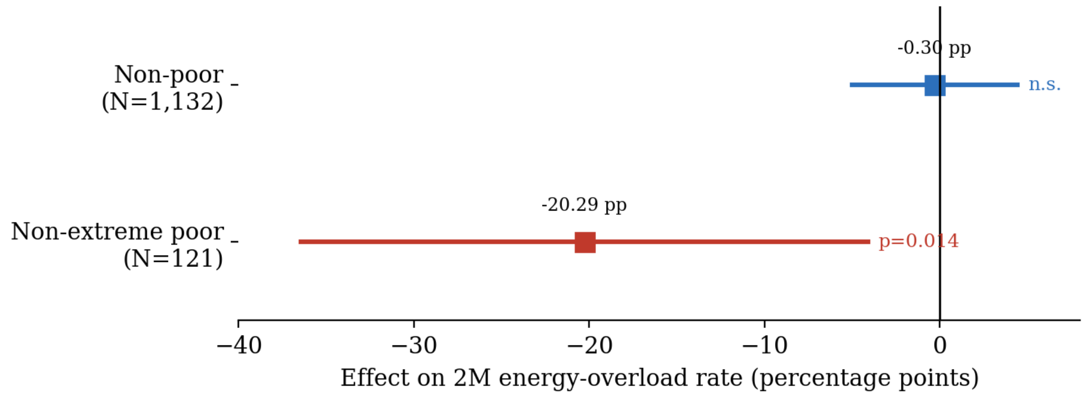

Figure 4: Heterogeneity of the FOSE effect on energy overload by poverty status (CEM, ±20 kWh). Source: ERCUE 2025.

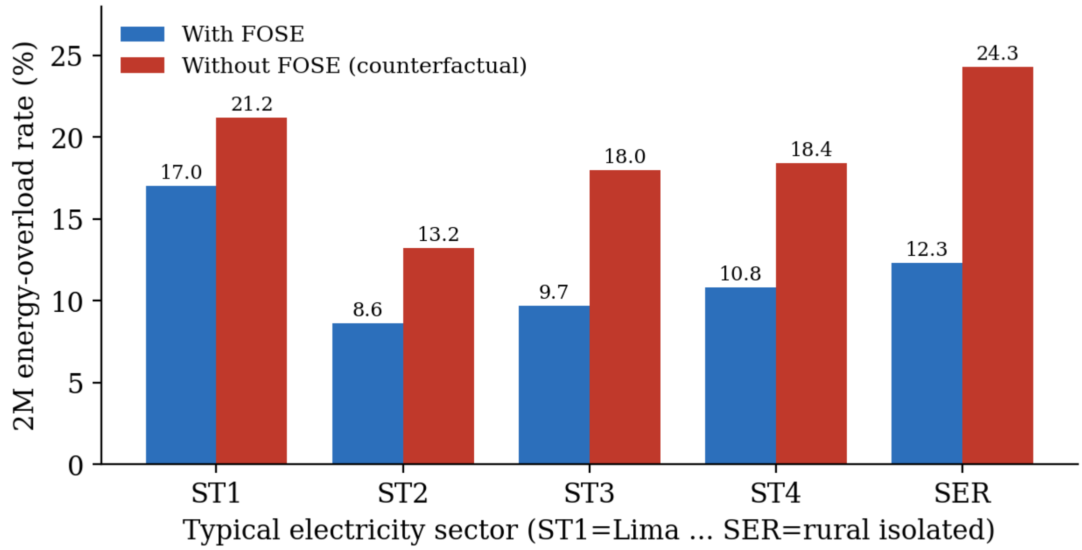

Figure 5: Counterfactual simulation: energy overload (2M) with and without the FOSE, by typical sector. Source: ERCUE 2025; eligible households.

threshold does possess discriminating power, but only among households that genuinely face energy vulnerability; because 69.4% of eligibles are not poor, the signal dilutes in the full sample. This −20 pp result is stable across all windows from ±15 to ±30 kWh.

4.4 Counterfactual: a scenario without the FOSE

To gauge the magnitude of the FOSE's protective effect, I construct a counterfactual that removes the discount (Figure 5). The energy overload rate would rise from 17.0% to 21.2% in ST1 (+4.3 pp), from 9.7% to 18.0% in ST3 (+8.3 pp), and from 12.3% to 24.3% in SER (+12.0 pp). The protective effect is largest in rural sectors, where the proportional discount of the first block (60%) has greater incidence. Thus, even though the instrument is poorly targeted in the aggregate, it provides meaningful protection precisely where consumption is lowest.

5 Conclusions and Policy Implications

This paper has evaluated the targeting and effectiveness of the FOSE as an energy-poverty mitigation instrument in Peru, using ERCUE 2025 microdata and quasi-experimental matching methods. Three conclusions follow.

First, the FOSE shows structural targeting deficiencies. The 69.4% inclusion error is a direct consequence of a quasi-universal design based solely on electricity consumption—a weak proxy for socioeconomic vulnerability whose discriminating power varies by sector (74.8–78.6% in urban areas versus 60% in rural areas). The main exclusion criterion in the SER is normative—the shared-meter rule—not the consumption threshold.

Second, the 140 kWh threshold produces no significant separation in the probability of energy overload between comparable eligible and excluded households. The null aggregate result is explained by the composition of the eligible group: 69.4% are not poor. Among non-extreme poor households the ATT is −20.29 pp ($p = 0.014$). The threshold has discriminating power, but it is masked by the massive inclusion of non-vulnerable households. The counterfactual shows that removing the FOSE would raise the overload rate by 4.3 pp (ST1) to 12.0 pp (SER).

Third, HEP is exactly zero in the analysis window: near the threshold, energy poverty is financial overload, not deprivation. The FOSE faces two distinct problems in distinct segments—deprivation among very-low-consumption households (where the barrier is access) and overload among intermediate-consumption households (where the barrier is affordability)—and a single consumption threshold cannot address both at once.

In short, the FOSE does not fail because the subsidy is insufficient, but because the criterion identifying the beneficiary is weakly correlated with the dimension of vulnerability it intends to correct. The results suggest three reform avenues: (i) a transition toward socioeconomic targeting using Peru's Household Targeting System (SISFOH), following the 2017 reform of Spain's *Bono Social*; (ii) an urgent review of the multi-meter exclusion rule; and (iii) a recalibration of the block structure or its complementation with direct transfers. The main limitation is the cross-sectional design; the natural extension is a difference-in-differences design exploiting the three ERCUE editions (2019, 2023 and 2025).

References

Arze del Granado, J., Coady, D., & Gillingham, R. (2012). The unequal benefits of fuel subsidies: A review of evidence for developing countries. *World Development, 40*(11), 2234–2248.

Bagnoli, L., & Bertoméu-Sánchez, S. (2022). How effective has the electricity social rate been in reducing energy poverty in Spain? *Energy Economics, 106*, 105792.

Boardman, B. (1991). *Fuel poverty: From cold homes to affordable warmth.* Belhaven Press.

Brown, C., Ravallion, M., & van de Walle, D. (2018). A poor means test? Econometric targeting in Africa. *Journal of Development Economics, 134*, 109–124.

Coady, D., Grosh, M., & Hoddinott, J. (2004). Targeting outcomes redux. *The World Bank Research Observer, 19*(1), 61–85.

EPOV (2023). *Energy poverty in the EU.* EU Energy Poverty Observatory, European Commission.

García Álvarez, G., & Tol, R. S. J. (2021). The impact of the Bono Social de Electricidad on energy poverty in Spain. *Energy Economics, 103*, 105554.

Iacus, S. M., King, G., & Porro, G. (2012). Causal inference without balance checking: Coarsened exact matching. *Political Analysis, 20*(1), 1–24.

Levinson, A., & Silva, E. (2022). The electric Gini: Income redistribution through energy prices. *American Economic Journal: Economic Policy, 14*(2), 341–365.

Llorca, M., & Rodríguez-Álvarez, A. (2025). Is Spain's energy voucher lighting the way for the poor? A microeconomic evaluation of the Bono Social Eléctrico. *Cambridge Working Papers in Economics 2542*, University of Cambridge.

Osinergmin (2025). *Reporte de análisis económico sectorial: Electricidad.* Gerencia de Políticas y Análisis Económico.

Thomson, H., Snell, C., & Liddell, C. (2016). Fuel poverty in the European Union: A concept in need of definition? *People, Place and Policy, 10*(1), 5–24.

Whittington, D., Nauges, C., Fuente, D., & Wu, X. (2015). A diagnostic tool for estimating the incidence of subsidies delivered by water utilities in low- and medium-income countries. *Utilities Policy, 34*, 70–81.